

Functional characterization of the transcription regulator WhiB1 from *Gordonia* sp. IITR100

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Abstract

WhiB is a transcription regulator which has been reported to be involved in the regulation of cell morphogenesis, cell division, antibiotic resistance, stress, etc., in several members of the family *Actinomycetes*. The present study describes functional characterization of a WhiB family protein, WhiB1 (protein ID: WP_065632651.1), from *Gordonia* sp. IITR100. We demonstrate that WhiB1 affects chromosome segregation and cell morphology in recombinant *Escherichia coli*, *Gordonia* sp. IITR100 as well as in *Rhodococcus erythropolis*. Multiple sequence alignment suggests that WhiB1 is a conserved protein among members of the family *Actinomycetes*. It has been reported that overexpression of WhiB1 leads to repression of the biodesulfurization operon in recombinant *E. coli*, *Gordonia* sp. IITR100 and *R. erythropolis*. A WhiB1-mut containing a point mutation Q116A in the DNA binding domain of WhiB1 led to partial alleviation of repression of the biodesulfurization operon. We show for the first time that the WhiB family protein WhiB1 is also involved in repression of the biodesulfurization operon by directly binding to the *dsz* promoter DNA.

INTRODUCTION

WhiB comprises a sub-family of *whi* genes and is so named because mutation in these genes turned the colour of *Streptomyces coelicolor* to white [1, 2]. WhiB family proteins have diverse regulatory effects on the cell. They are transcriptional regulators involved in various important functions such as cell division, pathogenesis, stress response, starvation, antibiotic resistance and oxidative stress [3–6].

In *Mycobacterium*, seven WhiB-like proteins (WhiB1–7) have been identified and their function deduced [7–9]. WhiB-like transcription factors are found in *Actinomycetes* but not in other bacterial orders [10]. WhiB of *Mycobacterium* contains conserved cysteine residues. The central two cysteines compose the CXXC motif by which it forms a complex with a [4Fe–4S] cluster that helps WhiB to bind with DNA, thereby making WhiB a DNA binding protein [11, 12]. In *Streptomyces*, 14 *whiB*-like genes are present and are involved in sporulation, morphological development, antibiotic biosynthesis, etc. [13].

To date, however, no study describing the exact role of WhiB has been reported in other members of the family *Actinomycetes* such as *Gordonia* spp. Despite regulating such important

cellular processes, very few studies have reported on the role of WhiB in different members of the *Actinobacteria*, with the WhiB of *Mycobacterium* and *Streptomyces* being an exception. The regulation by WhiB family proteins also remains poorly explored.

Gordonia sp. IITR100 is a Gram-positive bacterium that belongs to the class *Actinobacteria*. This bacterium is mostly known for its ability to metabolize organic sulphur compounds. The desulfurization genes are present as an operon driven by a 385 bp *dsz* promoter. Recently, protein(s) that bind to the *dsz* promoter were isolated and identified by an *in vivo* assay in which the cells were grown in media containing sodium sulphate as a sulphur source. Since the promoter is repressed in the presence of inorganic sulphur, it is likely that the repressor proteins bind to the promoter to repress the activity of the operon. In this study, the WhiB1 protein was identified in the assay as one of the proteins that was bound to the *dsz* promoter [14]. As a control experiment, an *in vivo* assay was also performed using primers for the kanamycin promoter. Because WhiB1 was not detected in the control experiment, it was suggested that the binding of WhiB1 was specific to the *dsz* promoter. Therefore, it is

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Abbreviations: DBT, Dibenzothiophene; EMSA, electrophoretic mobility shift assay; HBP, Hydroxybiphenyl; RLU, relative light unit; RT-PCR, reverse transcriptase PCR.

Table 1. Strains and plasmids used in the present study

Strain or plasmid	Relevant features	Source
<i>Escherichia coli</i> DH5α	Cloning strain	Invitrogen
BL 21 (DE3) pLysS	Expression strain	Invitrogen
<i>Gordonia</i> sp. IITR100	Completely sequenced, 5.6 Mb	Laboratory isolate (MCC2877)
pET29a	Expression vector, 5.3 kb, Kan ^R , pBR322 ori, T7 promoter	Novagen, USA
pRSG43	Kan ^R , 5.2 kb, contains PRC4 replicon	Gift of Dr Shavandi [26]
pHYBP109	Amp ^R , contains <i>luxAB</i> genes	Gift of Dr Victor De Lorenzo
pTACG	Expression vector, 9.2 kb, Kan ^R , <i>tac</i> promoter	PhD thesis, Pooja Singh, 2015
pET-WhiB	Kan ^R , 5.4 kb, WhiB1 gene cloned between <i>NdeI</i> and <i>HindIII</i> restriction sites in pET29a vector	This study
pET-WhiB1_Mut	Kan ^R , 5.4 kb, mutant WhiB1 gene cloned between <i>NdeI</i> and <i>HindIII</i> restriction sites in pET29a vector	This study
pTAC-WhiB	Kan ^R , 5.4 kb, WhiB1 gene cloned between <i>NdeI</i> and <i>HindIII</i> restriction sites under <i>tac</i> promoter	This study
pRSG43-WhiB	Kan ^R , 6.8 kb, WhiB1 gene under <i>tac</i> promoter is cloned between <i>BspHI</i> restriction sites	This study

likely that the protein plays a role in regulation of the *dsz* operon.

In this study, we show that WhiB1 protein from *Gordonia* sp. IITR100 is involved in various important functions such as regulation of the desulfurization operon, cell morphology and chromosome segregation.

METHODS

Bacterial strains and plasmids

The bacterial strains and plasmids used in the present study are given in Table 1.

Media and growth conditions

Gordonia sp. IITR100 was grown in minimal medium containing sodium sulphate (3 mM) as a sulphur source. The minimal salt medium contained (per litre) Na₂HPO₄ (2.0 g), KH₂PO₄ (1.0 g), ammonium oxalate (4.25 g), MgCl₂ (0.4 g), sucrose (50 mM) and trace elements [14].

LB medium was used for growth of *E. coli* cells in all studies except for the luciferase assay for which M9 medium was used. M9 medium contained (per litre) Na₂HPO₄ (6.0 g), KH₂PO₄ (3.0 g), ammonium chloride (1 g), NaCl (1 g), MgCl₂ (1 mM), CaCl₂ (1 mM), glucose 0.4% and trace elements [trace elements consisted of, per litre: KI 0.05 g, LiCl 0.05 g, MnCl₂·4H₂O 0.8 g, H₃BO₃ 0.5 g, ZnCl₂ 0.1 g, CoCl₂·6H₂O 0.1 g, NiCl₂·6H₂O 0.1 g, BaCl₂ 0.05 g, (NH₄)₆Mo₇O₂₄·2H₂O 0.05 g, SnCl₄·2H₂O 0.5 g, Al(OH)₃ 0.1 g]. M9 medium containing Dibenzothiophene (DBT) (0.1 mM) as a sulphur source was used for the luciferase assay. Cells were grown at 37 °C [14].

The antibiotics used in the medium were kanamycin (50 µg ml⁻¹), ampicillin (100 µg ml⁻¹) and spectinomycin (40 µg ml⁻¹). All the chemicals were of molecular grade.

Cloning and expression of WhiB1 in pET29a vector

Genomic DNA was isolated from *Gordonia* sp. IITR100 following the protocol of Singh *et al.* [15]. The gene coding for the protein WhiB1 was PCR-amplified from the isolated genomic DNA with the primers containing restriction sites for *NdeI* and *HindIII* (Table 2). The digested PCR product was cloned under control of the T7 promoter in the plasmid pET29a (digested with the same restriction enzymes). The clone was confirmed by insert release. The recombinant plasmid was transformed to different expression strains to overexpress the protein of interest.

Table 2. Primer name sequence (5'→3')

Primer name	Sequence(5'→3')
WhiB-F	CTAGCATATGATGGCTTTGGAAA ACCCGATCGGCAATCCG
WhiB-R	CTAGAAAGCTTGATGATGCCGCGCT TGAGACGCCGCGCTC
qdszC-F	GCCTTCATTGTGCGCTTCAT
qdszC-R	GCGCGATCCCTAAATAGACG
qTetR-F	GATCGATGCGAAAGGCAGAT
qTetR-R	CCTTGGAGTCGAAATGGTGATAG
Gyrase-F	CTGCCGTGAGTGAGTACCCA
Gyrase-R	AACCTGAACCGCACCAACC
qWhiB-F	CAATCCGCTTGAACACAGA
qWhiB-R	GCACCAACGAAAGATGACGA
Mutant FP	GCCAAGCGGATCTGCGCTGGCTGCGAGGTCAAGTCC
Mutant RP	GGACTTGACCTCGCAGCCAGCGCAGATCCGCTTGGC

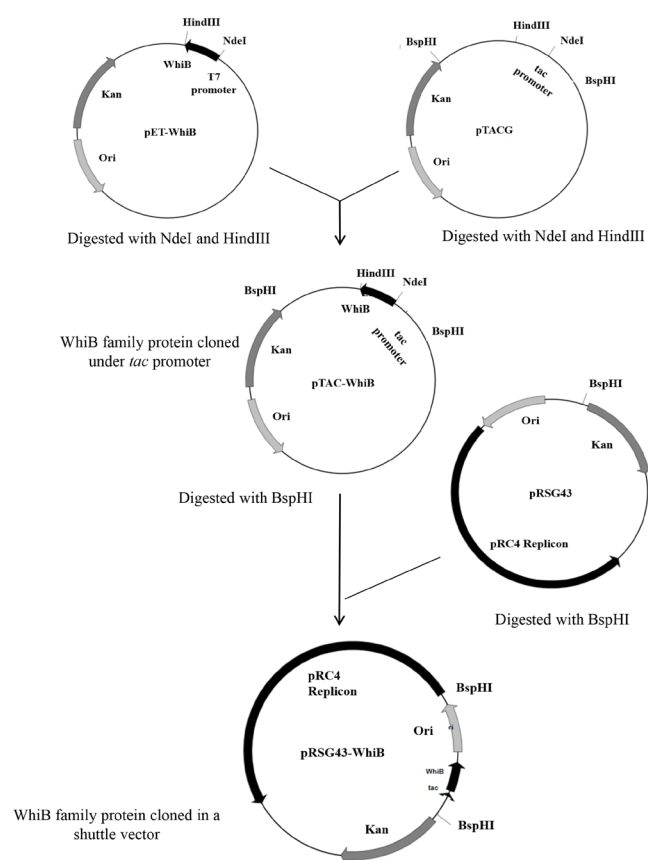


Fig. 1. Construction of a shuttle vector containing the gene for WhiB1 protein under the *tac* promoter.

All recombinant DNA methods were performed following the protocols given by Sambrook and Russel [16].

Cloning and expression of WhiB mutant in pET29a vector

Genomic DNA from *Gordonia* sp. IITR100 was isolated and used as the template for overlap extension PCR for site-directed mutagenesis of the gene for WhiB1. The codon encoding the 116-Gln in WhiB_{WT} protein was replaced by 116-Ala in the WhiB1_Mut. Two sets of primers, i.e. Set 1 containing WhiB FP and Whi_Mut RP, and Set 2 containing WhiB_Mut FP and WhiB RP, were used to obtain the amplicons W1-363bp and W2-132bp, respectively, which had an overlap of 36 bp containing the required mutation. The amplicons W1 and W2 were used to amplify *whiB1_Mut* using primers WhiB FP and WhiB RP containing restriction sites for *NdeI* and *HindIII*. The digested PCR product was cloned under control of the T7 promoter in the plasmid pET29a, which was digested with the same restriction enzymes. The clone was confirmed by insert release and the mutation was confirmed by sequencing. The plasmid harbouring the mutated gene *whiB1* was transformed to *E. coli* BL21(DE3)-pLysS expression strains for overexpression analysis.

Cloning of the WhiB1 protein into the pTAC vector

The gene for the protein WhiB1 was also cloned under the *tac* promoter between the restriction sites *NdeI* and *HindIII* in plasmid pTAC. The clone (pTAC-WhiB) was confirmed by insert release by restriction digestion. The overexpression of WhiB1 in BL21 (DE3) pLysS was analysed by SDS-PAGE.

Construction of shuttle vector for expression of WhiB1 in *Gordonia* and *Rhodococcus*

A shuttle vector for expression of the WhiB1 in *Gordonia* and *Rhodococcus* was constructed. For this, the pRC4 replicon was obtained by digesting plasmid pRSG43 with restriction enzyme *BspHI*. A fragment of ~5 kb in size was gel-eluted. The WhiB1 gene along with *tac* promoter was obtained by digesting plasmid pTAC-WhiB with the same restriction enzyme *BspHI*. The small fragment was gel-eluted. The eluted fragments were ligated to generate a construct pRSG43-WhiB, which can replicate in both *E. coli* and *Gordonia/Rhodococcus*.

Preparation of electrocompetent cells of *Gordonia* and *Rhodococcus*

Competent cells of *Gordonia* sp. IITR100 were prepared according to the protocol given by Singh *et al.* [17]. To prepare competent cells of *Rhodococcus erythropolis* IGTS8, a single colony of the bacterium was inoculated in 100 ml LB medium and allowed to grow at 30 °C until an OD₆₀₀ of 0.5 was reached. Cells were then harvested by centrifugation at 3500 r.p.m. for 10 min followed by washing with ice cold 10% glycerol twice. Cells were then resuspended in 1 ml of 10% glycerol and aliquots of 100 µl were stored at –80 °C until further use.

Luciferase assay

To study the role of WhiB1 and WhiB1_mut in biodesulfurization, a luciferase assay was performed using the Dual-Luciferase Reporter Assay System (Promega). Briefly, three plasmids, pTB2 (which contains the *dsz* operon), pHYBP109 (which contains *luxAB* and *hbpR* present under the *Phbpc* promoter) and pET-WhiB/pET-WhiB1 Mut were transformed in *E. coli* cells. *E. coli* cells harbouring these plasmids were grown in M9 media containing DBT (0.1 mM) as a sulphur source until an OD₆₀₀ of ~1. The cells were further harvested by centrifugation (12000 r.p.m. for 2 min) and finally resuspended in 720 µl resuspension buffer, which consisted of 50 mM glucose, 25 mM Tris-HCl (pH 8) and 10 mM EDTA (pH 8). Lysozyme (3 µl from 45 mg ml⁻¹ stock) and 1 µl RNase (from 10 mg ml⁻¹ stock) were added to the above suspension and incubated at 37 °C for 1 h. The cell lysate was centrifuged at 12000 r.p.m. for 10 min after incubating the lysed cells with PMSF (final concentration 1 mM) in ice for 30 min, and the supernatant was collected in a fresh tube. On adding the substrate (30 µl luciferin), to an aliquot of the supernatant (70 µl), the light produced was detected by a luminometer. The delay time was kept constant for all the samples (~2 s) and the light was measured for 10 s after which the final value in relative light units (RLU) s⁻¹ was recorded [14].

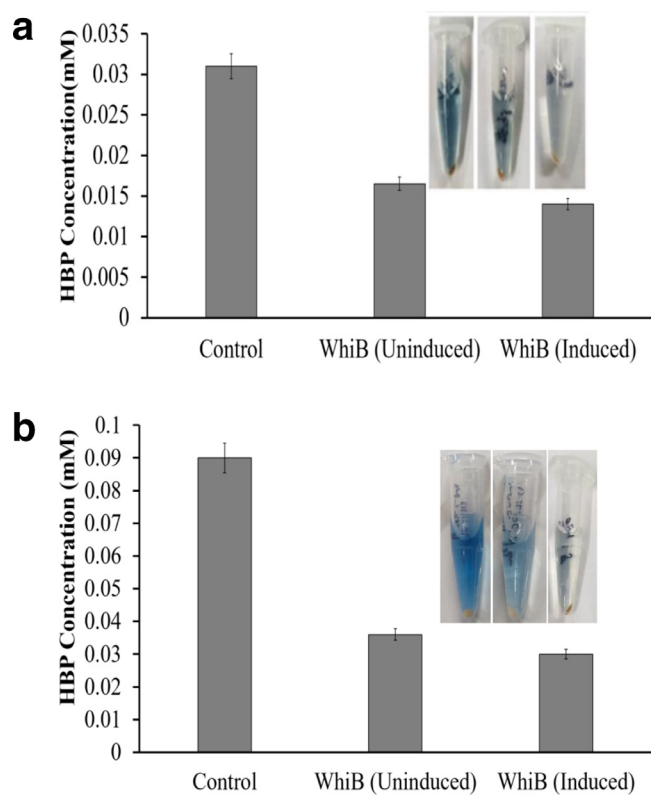


Fig. 2. Graph showing the effect of WhiB1 on activity of the *dsz* operon in (a) recombinant *Gordonia sp. IITR100* and (b) recombinant *Rhodococcus erythropolis IGTS8*. A decrease in activity of the operon was observed in the presence of WhiB1 protein. This further decreased when WhiB1 protein is overexpressed. All the experiments were performed in triplicate and were repeated three times.

E. coli cells harbouring the plasmids pTB2 and pPHYBP109 were used as a control for the luciferase assay.

Gibbs assay

A Gibbs assay was used to monitor the effect of the WhiB1 protein on the biodesulfurization activity in *Gordonia sp. IITR100* and *R. erythropolis*. The pH of 1 ml of culture was adjusted to 8 with 10% (w/v) sodium carbonate. Gibbs reagent (10 µl from 1% 2,6 dichloroquinone-4-chloroimide in ethanol) was added to the culture and incubated at 30 °C for 30 min in the dark. The cell suspension was centrifuged at 10000 r.p.m. for 2 min to remove the cells and absorbance of the supernatant was determined at OD₆₁₀. Cells containing the empty vector pRSG43 was used as a control.

Purification of WhiB1 protein

To validate the binding of WhiB1 protein to the *dsz* promoter, the protein was purified and an electrophoretic mobility shift assay (EMSA) was performed. For purification, a single colony of BL21 (DE3) pLysS cells harbouring the plasmid containing the gene coding for the WhiB1 protein was inoculated in 200 ml of LB medium and induced for overexpression with 1 mM IPTG. After 5 h of induction,

the cells were harvested by centrifugation at 3500 r.p.m. for 10 min. The cells were resuspended in 4 ml denaturing buffer A (100 mM Na₂PO₄, 10 mM Tris, 8 M urea, pH 8) and sonicated on a Q125 sonicator (Qsonica sonicators) by giving 30 s bursts and 20 s rest for 5 min at an amplitude of 25%. The cell debris was removed by centrifugation at 10000 r.p.m. for 15 min and the supernatant was collected in a fresh tube. The sonicated supernatant was added to the Ni-NTA column equilibrated with buffer A. The suspension was incubated at 4 °C on a rotating wheel for 2 h to allow binding of the protein to Ni-NTA agarose resins. The flow-through was collected and the column was washed with buffer B (100 mM Na₂PO₄, 10 mM Tris, 8 M urea, pH 6.5). The proteins bound to the column were eluted by using elution buffers C and D (100 mM Na₂PO₄, 10 mM Tris, 8 M urea, pH 5.5; and 100 mM Na₂PO₄, 10 mM Tris, 8 M urea, pH 4.5, respectively). The different fractions collected were run on a 12% SDS-PAGE gel to check the fraction containing the protein of interest. The elution fraction with buffer C (containing the WhiB1 protein) was renatured by adding the eluate (5 ml) dropwise to 50 ml renaturation buffer (10% glycerol, 25 mM Tris-Cl and 150 mM NaCl, pH 8) in a cold room by stirring continuously with a magnetic stirrer. The renatured protein solution was further dialysed overnight against the renaturation buffer with two or three changes to remove traces of urea. The renatured sample was run on a 12% SDS-PAGE gel to check the purity of the protein.

EMSA with purified WhiB1

An EMSA was performed to determine the binding of WhiB1 with the *dsz* promoter. For this, the *dsz* promoter was Cy5-labelled by PCR amplification using Cy5 PS19 (5' CAGTCATATGCGCGTATGTGTCCTCTAACCCTAA ATAGCG 3') and PS18 (GAGCAGATCTGGCCATGATCGACCGCTCGTCCATCACGC) primers. The reaction mixture consisting of Tris binding buffer (20 mM Tris-Cl, 5 mM MgCl₂, 0.1 mM EDTA, 6% sucrose, 100 mM KCl and 1 mM DTT, pH 7.5), poly d(I-C), poly-L-lysine, 12 ng of labelled promoter and increasing concentrations of purified protein (10–40 µg) was incubated for 4 h at 4 °C and was then run on a 6% native TBE gel for 2 h at 100 V. The gel was then scanned in a fluorimager to visualize the shift. As a control, all components of the reaction mixture mentioned above except the protein were used.

Cell morphology and chromosome segregation studies

To study the effect of WhiB1 on cell morphology and chromosome segregation, *E. coli* cells harbouring the plasmid pET-WhiB were inoculated and the samples were collected at different time intervals. To study the effect of WhiB1 on cell shape and length, 1 ml of the sample was harvested by centrifugation at 2500 r.p.m. for 5 min, washed twice with 50 mM HEPES buffer (pH 7.5) and finally resuspended in 100 µl of the same buffer. About 2 µl of the washed cells was

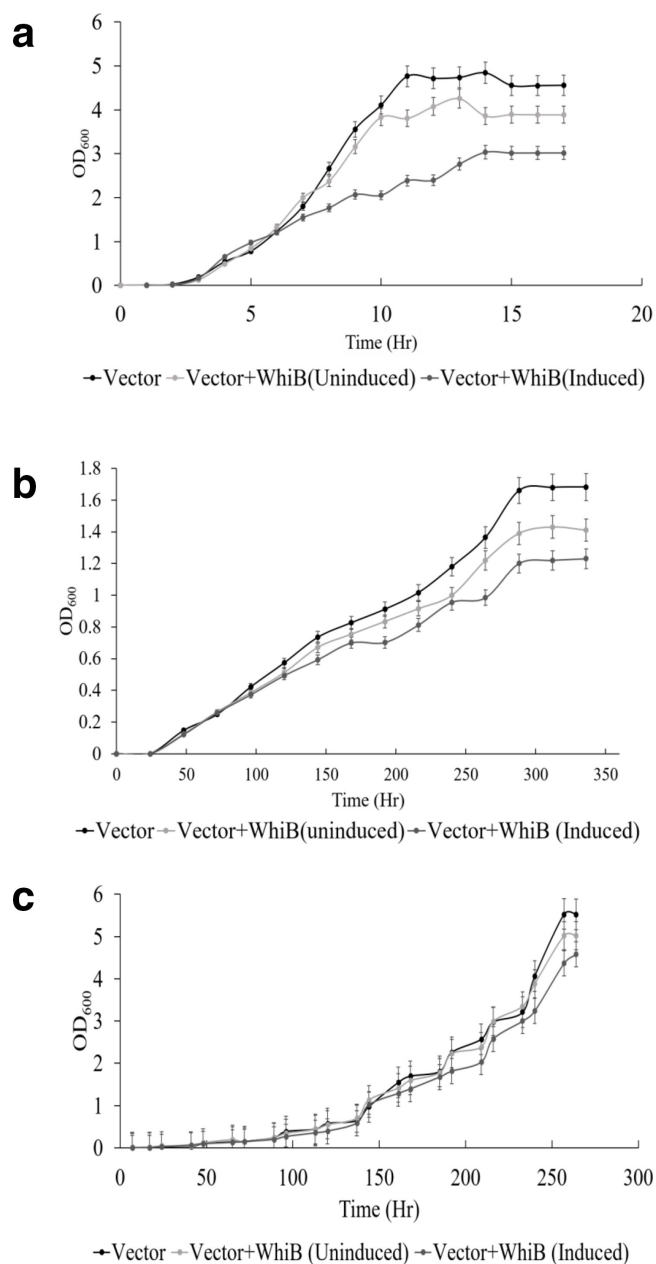


Fig. 3. Graphs showing the effect of WhiB1 on growth in (a) recombinant *E. coli*, (b) in recombinant *Gordonia* sp. IITR100 and (c) in recombinant *Rhodococcus erythropolis* IGTS8. A decrease in growth of the bacteria was observed in the presence of WhiB1 protein. All the experiments were performed in triplicate and were repeated twice.

placed on a glass slide and observed under a microscope for observation of differences in cell morphology [18].

To study the effect of WhiB1 on chromosome segregation, the samples were prepared as mentioned above followed by staining of the nucleoid with DAPI. DAPI (5 µl, concentration 10 µg ml⁻¹) was added to the cells and kept for 2 h in the dark. About 2 µl of the stained cells was placed on a

glass slide and observed under the microscope to observe differences in chromosome segregation [18].

Cells containing the empty vector pET29a were used as a control.

RT-PCR

To determine the effect of WhiB1 protein on the *dsz* operon gene at the transcript level, reverse transcriptase PCR (RT-PCR) was performed. For this, RNA was isolated from *Gordonia* sp. IITR100 containing the shuttle vector expressing WhiB1 induced with 1 mM IPTG. The isolated RNA was converted to cDNA using a cDNA synthesis kit (Thermo Fischer Scientific). The RT-PCR reaction was set up using the above prepared cDNA as a template and primers for the intragenic region of *dszC*, TetR family protein (TFR) and *whiB1* (Table 2). The fold change in the expression of *dsz* genes was calculated via the double delta C_T method. As a control experiment, *Gordonia* sp. IITR100 cells harbouring empty vector were used. The housekeeping gene *gyrA* was used as an experimental control.

Bioinformatics studies

The three-dimensional structure of the WhiB1 family protein was elucidated computationally using expasy software (<https://swissmodel.expasy.org/>) based on similarity with protein 5OAY.1 (transcriptional regulator WhiB1 of *Mycobacterium tuberculosis*, 44.9% similarity).

Multiple sequence alignment of this protein was done with other bacteria using CLUSTAL omega (<https://www.ebi.ac.uk/Tools/msa/clustalo/>).

RESULTS

Cloning of the gene coding for the WhiB1 protein

The gene coding for the WhiB1 protein identified previously by an *in vivo* assay [19] was cloned under control of the *tac* promoter and the vector constructed was named pTAC-WhiB. The promoter along with the gene coding for WhiB1 protein from pTAC-WhiB was cloned in a shuttle vector containing the pRC4 replicon for replication in *Gordonia/R. erythropolis* and the constructed vector was named pRSG43-WhiB (Fig. 1).

WhiB1 reduces activity of the biodesulfurization operon

To study the effect of overexpression of WhiB1 in its native host, the shuttle plasmid (pRSG43-WhiB) was transformed to *Gordonia* sp. IITR100. Recombinant *Gordonia* sp. IITR100 was grown in minimal medium containing DBT as the sulphur source and biodesulfurization activity was measured by the Gibbs assay, which provides an indication of the amount of 2-Hydroxybiphenyl (HBP) produced from DBT. This corresponds to the activity of the *dsz* operon. The formation of 2-HBP decreased from 0.03 mM (control) to about 0.018 mM (~40% decrease) in the presence of WhiB1. A further decrease in 2-HBP production was observed

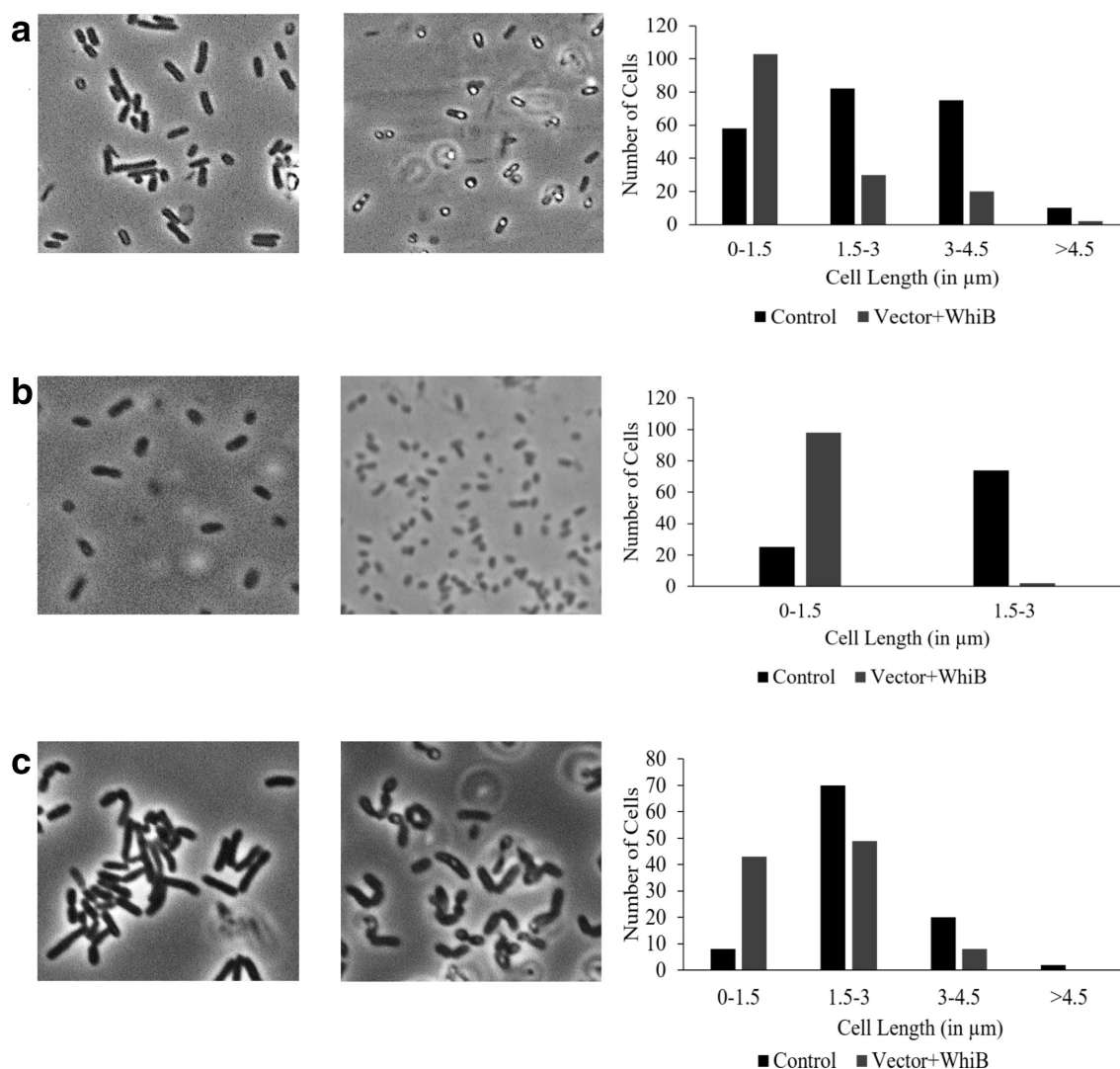


Fig. 4. Effect of WhiB1 on cell morphology, chromosome segregation and cell length. The left lane shows control cells (cells containing empty vector), the right lane shows cells containing WhiB1 supplied through the plasmid and the rightmost lane is graphs showing the decrease in the number of elongated cells in the presence of WhiB1 in (a) *E. coli* cells, (b) *Gordonia* sp. IITR100 and (c) *Rhodococcus erythropolis* IGTS8. A decrease in cell length is seen in cells where WhiB1 is present as compared to the control. The majority of cells tend to be short rod-shaped or coccoid rather than elongated in the presence of WhiB1 protein. Uneven chromosomal segregation is seen where WhiB1 is present as compared to the control. The majority of cells tend to be anucleate in the presence of WhiB1. All the experiments were performed in triplicate and were repeated twice.

on induction of the WhiB1 protein (Fig. 2a). A similar observation was found in recombinant *R. erythropolis* IGTS8 containing the gene for the WhiB1 protein supplied *in trans*. In control cells containing the empty pRSG43 plasmid, the concentration of 2 HBP produced was nearly 0.09 mM whereas the amount was reduced to approximately 0.035 mM (~40% decrease) when the gene for WhiB1 was supplied *in trans*. A further decrease in 2-HBP production was observed on induction of the WhiB1 (Fig. 2b). The results suggest that the WhiB1 protein represses the activity of the *dsz* operon in recombinant *Gordonia* and *R. erythropolis* IGTS8.

WhiB family proteins have been shown to be involved in cell morphogenesis and growth [10].

WhiB1 affects the cell growth of bacteria

To study the effect of WhiB1 on cell growth, BL21 (DE3) pLysS cells harbouring the pET-WhiB plasmid were inoculated. When an OD_{600} of ~0.5 was reached, the culture was divided and one portion was induced while the other was uninduced. Samples were collected at regular intervals and the OD_{600} was monitored. As a control, BL21 (DE3) pLysS cells harbouring pET29a plasmid were used. A decrease in cell growth was observed in the presence of WhiB1. Growth

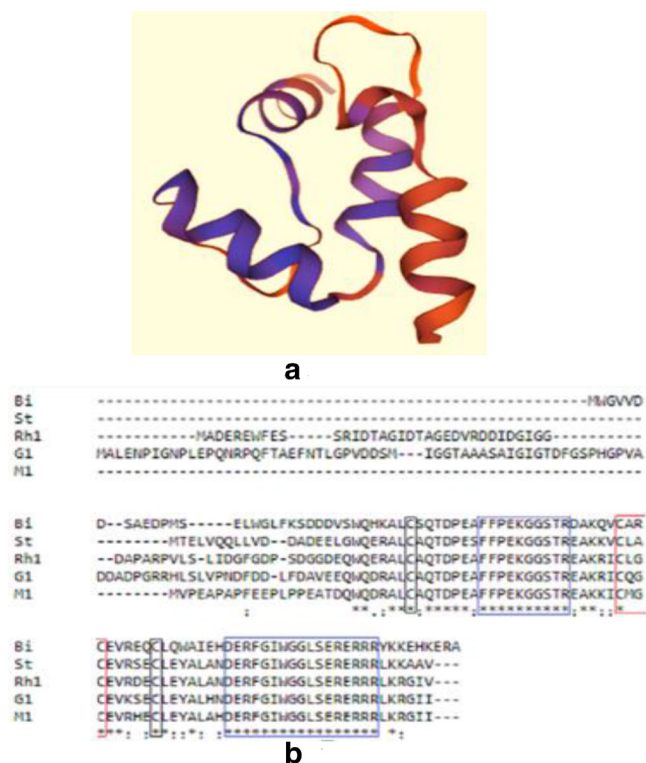


Fig. 5. (a) A three-dimensional structure of the WhiB1 protein constructed based on the structure of transcription regulator WhiB (5OAY.1.A, 44.93%) of *Mycobacterium tuberculosis*. (b) Multiple sequence alignment of the WhiB protein in five different organisms (*Bifidobacterium bifidum*, *Streptomyces coelicolor*, *Rhodococcus sp.*, *Gordonia sp.* IITR100 and *Mycobacterium tuberculosis*).

decreased further in cells overexpressing WhiB1 (Fig. 3a). A similar growth pattern was observed in recombinant *Gordonia sp.* IITR100, which contained WhiB1 supplied *in trans* (Fig. 3b) and recombinant *R. erythropolis* (Fig. 3c) when compared to the control.

WhiB1 affects the cell morphology and chromosome segregation of bacteria

The effect of WhiB1 on cell morphology was studied by microscopy. For this, *E. coli* cells containing pET-WhiB were inoculated, induced with IPTG when an OD₆₀₀ of ~0.5 was reached, and samples were prepared for microscopy analysis. As a control, cells containing empty vector were used. In the presence of WhiB1 protein, the cells tended to be cocci in shape whereas in cells where WhiB1 was absent the cells tended to be more rod-shaped (Fig. 4a). Furthermore, there was a significant change in cell length. The elongated *E. coli* cells were shortened in the presence of WhiB1. The cell length of the majority of the cells decreased from 3.4 to 2.5 µm in the presence of WhiB1. Therefore, we can say that WhiB1 affects cell shape and length. This result is not surprising as many studies have reported the involvement of the Whi family of proteins in bringing about morphological changes [10]. A

similar effect of WhiB1 on cell length and morphology was observed in *Gordonia sp.* IITR100 (Fig. 4b) and *R. erythropolis* IGTS8 (Fig. 4c).

To study the effect of WhiB1 on chromosome segregation, the nucleoid of *E. coli* cells containing WhiB1 was stained with DAPI and the changes were monitored through microscopy. Similarly, cells containing the empty vector were prepared and observed under the microscope. Uneven chromosome segregation was observed in the cells containing WhiB1 supplied *in trans*. Many cells were found to be anucleate in the presence of WhiB1 (Fig. 4a). A similar effect of WhiB1 on cell morphology and chromosome segregation was found in members of the *Actinobacteria*.

To determine whether the decrease in biodesulfurization activity is due to decreased growth and affected cell morphology or to a decrease in the transcripts of the desulfurization operon, the structure of the WhiB1 protein was analysed.

WhiB1 is a conserved protein in bacteria of *Actinomycetes*

The three-dimensional structure of WhiB1 protein was elucidated computationally using expasy software. The protein has a helical structure (Fig. 5a). Multiple alignment of the WhiB protein from different members of the family *Actinomycetes* (*Gordonia* IITR100, *Streptomyces coelicolor*, *Bifidobacterium bifidum*, *Rhodococcus aetheriorans*, *Mycobacterium tuberculosis*) was performed using the CLUSTAL omega bioinformatics tool. The WhiB protein contains four conserved near-invariant cysteine residues C(×n)C(×2)C(×5)C, including the common CXXC motif (in the red box) and some conserved regions (in the blue box) (Fig. 5b). The central two cysteines form a CXXC motif by which it complexes with a [4Fe–4S] cluster that helps WhiB to bind with DNA.

WhiB1 binds to the *dsz* promoter

To determine the direct binding of WhiB1 protein with the *dsz* promoter, EMSA was performed. For this, *E. coli* BL21(DE3) pLysS cells harbouring the plasmid pET-WhiB1 were used. Maximum protein expression was found after 5 h of induction. SDS-PAGE showing expression of the transcriptional regulator WhiB1 is shown in Fig. 6a. The protein was further purified and the purified protein was visualized on an SDS PAGE gel (Fig. 6b).

To verify the binding of WhiB1 to the *dsz* promoter, an EMSA was performed with Cy5-labelled promoter and purified protein. A clear shift was observed in the presence of the protein (Fig. 6c), suggesting that the protein binds to the promoter DNA.

WhiB1 expression represses the transcription of *dsz* genes

WhiB1 overexpression affects cell growth, cell morphology and chromosome segregation. In addition, overexpression

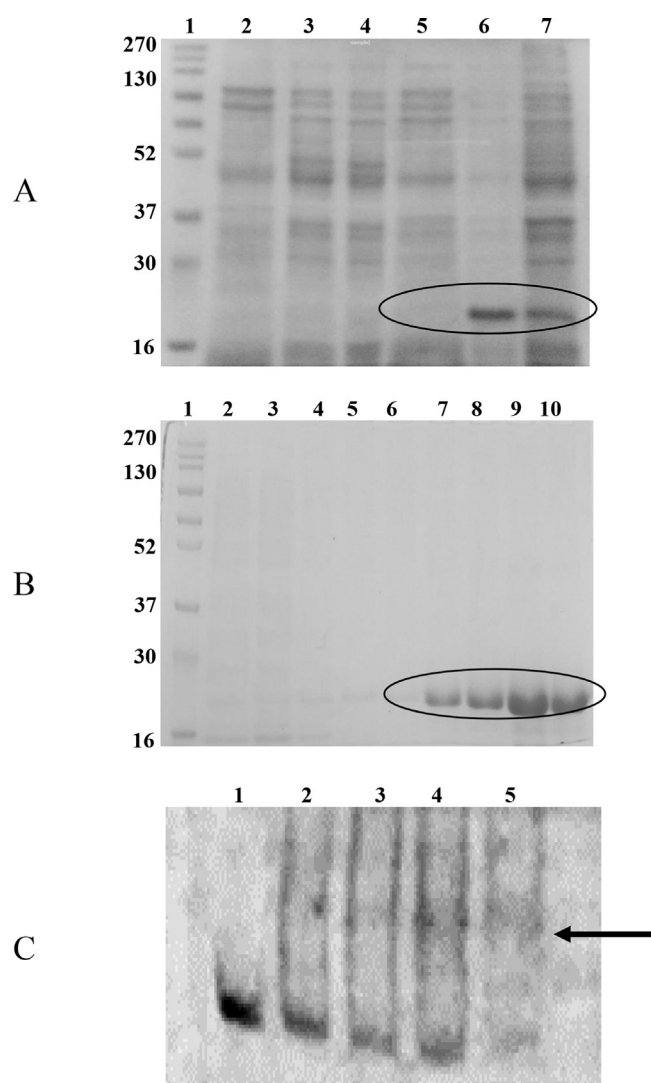


Fig. 6. SDS-PAGE showing expression, purification and EMSA of the WhiB1 protein. (a) Lane 1: marker, lane 2: uninduced sample [BL21 (DE3)], lane 3: 5 h induced sample [BL21 (DE3)], lane 4: overnight induced sample [BL21 (DE3)], lane 5: uninduced sample [BL21 (DE3) pLysS], lane 6: 5 h induced sample [BL21 (DE3) (pLysS)], lane 7: overnight induced sample [BL21 (DE3) (pLysS)]. (b) SDS-PAGE showing purification of overexpressed WhiB1. Lane 1: marker, lanes 2, 3: flow through, lanes 4–6: washing, lanes 7–10: elution fractions. (c) EMSA demonstrating binding of WhiB1 to the *dsz* promoter. Lane 1: control (only labelled DNA), lanes 2–5: Cy5-labelled *dsz* promoter + purified WhiB1 protein (10, 20, 30 and 40 μ g respectively). The experiments were performed in triplicate and were repeated three times.

of WhiB1 decreased the biodesulfurization activity. The decreased biodesulfurization activity could be due to this overall impact on cells due to WhiB1 overexpression or due to its action as a repressor. To determine if WhiB acts as a repressor of the *dsz* operon, the level of transcripts of various genes were determined with or without WhiB1. We found that with the overexpression of WhiB1, the transcript of the *dsz* gene (*dszC*) decreased significantly. We also observed that the transcript of the *dsz* operon activator, the

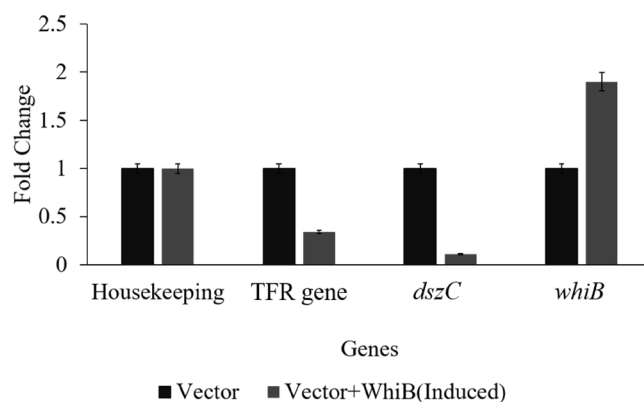


Fig. 7. Effect of WhiB1 on the transcript of the TetR family protein (TFR) and the *dsz* operon gene (*dszC*). A decrease in transcript abundance of the TetR family protein (TFR) and the *dsz* operon gene was seen with an increase in the number of *whiB* transcripts. All the experiments were performed in triplicate and were repeated three times.

TetR family protein (TFR protein), was reduced (Fig. 7). This suggests that WhiB1 can repress the operon either directly by binding to the *dsz* promoter or by inhibiting the effect of the *dsz* operon activator, the TFR protein.

A point mutation in the DNA binding domain of WhiB1 led to partial alleviation of repression

To determine the role of WhiB1 in repression of the biodesulfurization operon, we used *E. coli*, a heterologous host. To determine whether WhiB1 represses the operon by binding to the *dsz* promoter, a WhiB mutant (WhiB_mut) was constructed. A point mutation was introduced at position 116 in the protein. Thus, glutamine at position 116 was replaced by alanine (Fig. 8a). Expression of the mutant protein was analysed by SDS-PAGE (Fig. 8b). Expression of both wild type WhiB1 and the WhiB1_mut protein was comparable.

A luciferase assay was performed in recombinant *E. coli*. For this, cells were transformed with plasmids pTB2 (containing the *dsz* operon), pPHYBP109 (containing *luxAB* luciferase genes under control of the *Phbpc* promoter, which is regulated by the presence/absence of hydroxyl biphenyls) and pET-WhiB/pET-WhiB1_Mut. *Phbpc* is activated in response to 2-HBP, and thus the *lux* transcript fused to *Phbpc* is regulated in parallel to the concentration of 2-HBP [20]. Therefore, the values obtained in the luminometer are an indication of the change in concentration of HBP produced by the cells, thereby suggesting the activation/repression of the *dsz* operon.

With only pTB2 and pPHYBP109 plasmids, the luciferase activities in recombinant *E. coli* were about 294 RLU s^{-1} , whereas in the presence of the WhiB1 protein expression plasmid, the activities decreased to 203 RLU s^{-1} (~30% decrease; Fig. 8c). The results suggest that the WhiB1 protein serves as a repressor of the *dsz* operon. WhiB1_Mut, when supplied *in trans*, resulted in about 251 RLU s^{-1} (~15% decrease). The results suggest that WhiB1_Mut did not repress the desulfurization operon to the same extent as observed in case of the

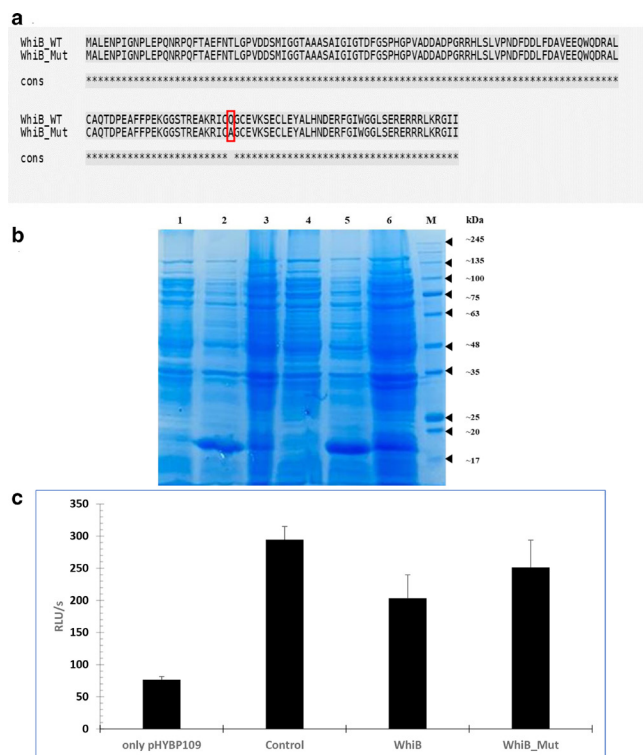


Fig. 8. Effect of mutation in the DNA binding domain of WhiB1. (a) The protein sequence of WhiB1 and WhiB1_mut. The mutation is highlighted in red. (b) Comparison of the expression of WhiB1 and WhiB1_mut on an SDS-PAGE. Lanes 1–3: WhiB_Mut (uninduced, 3 h and overnight after induction respectively), Lanes 4–6: WhiB_WT (uninduced, 3 h and overnight after induction respectively), Lane M: protein ladder (Blu elf, GeneDirex). (c) Biodesulfurization activity measured by a luciferase assay in the presence of WhiB1 and WhiB1_mut.

wild type WhiB1. Because the point mutation was introduced in the DNA binding domain of WhiB1, the results suggest that WhiB1 represses the *dsz* operon by binding to the promoter DNA. A single point mutation led to partial alleviation of repression.

DISCUSSION

A protein belonging to the WhiB family, WhiB1, was identified in an *in vivo* based assay with medium containing sodium sulphate. The binding of WhiB1 was further demonstrated by an EMSA with purified protein and the *dsz* promoter. WhiB-like proteins together with WhiA have been shown to bind to about 240 promoters regulating the genes for cell division and development [21, 22]. It is known that the proteins interacting with the promoter DNA may have a direct or indirect role in increasing or decreasing the transcription of genes downstream of the promoter [21, 22]. The protein when supplied *in trans* resulted in decreased desulfurization activity in recombinant *E. coli*, *Gordonia* and *R. erythropolis*. This led us to hypothesize that the protein is serving as a repressor of the promoter. Although to date there are no reports of WhiB1 regulating the *dsz* operon, the role of WhiB family

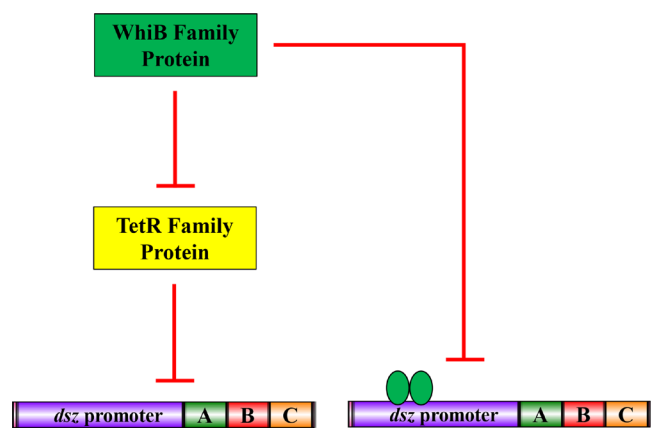


Fig. 9. Proposed mechanism of regulation of the *dsz* operon by WhiB1. The WhiB1 protein represses the *dsz* operon by either of the two proposed methods or a combination of the two. In the first mechanism, WhiB1 represses the transcription of the operon activator, TetR family protein, which in turn represses the operon. In the second mechanism, the protein binds to the *dsz* promoter and represses the operon.

proteins as a transcriptional regulator is not new [23–25]. It is interesting to note that although the WhiB proteins are unique to *Actinobacteria*, their effect upon overexpression was also observed in the heterologous host *E. coli*. A decrease in luciferase activity of about 30% was observed, suggesting that WhiB1 represses the operon in *E. coli*. When glutamine in the conserved CXXC motif in the DNA binding domain was mutated to alanine, a 15% decrease in activity was observed. Thus, a point mutation in the DNA binding domain of WhiB1 resulted in the partial alleviation of repression. It is likely that glutamine, which can form multiple hydrogen bonds, when replaced by alanine affected the binding of WhiB1_Mut to the DNA.

WhiB affects cell morphology and chromosome segregation in bacteria. Bush [25] and Chater [24] reported that the Whi family of proteins is responsible for spore formation in bacteria. It has also been reported that WhiB is essential for early morphological differentiation as well as for the regulation of initiation of sporulation [25].

Overexpression of WhiB-like protein has been previously shown to downregulate a TetR family regulator responsible for ACT (Actinorhodin) transport [13]. Similarly, in our case overexpression of WhiB1 resulted in reduced expression of TetR family transcriptional regulator and desulfurization genes. Based on the above observations, a mechanism of repression of the operon by WhiB1 is proposed in Fig. 9 where the WhiB1 protein can repress the operon by directly binding to the promoter or indirectly through the TetR family protein or both.

This is the first report on the role of the WhiB family protein from the bacterium *Gordonia* sp. IITR100. A protein belonging to the Whi family of transcription regulators was isolated, which was found to serve as a repressor of the *dsz*

operon. This protein also plays a role in chromosome segregation and cell morphology.

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Conflicts of interest

The authors declare that there no conflicts of interest.

Ethical statement

This article does not contain any studies with human participants or animals performed by any of the authors.

References

- Geiman DE, Raghunand TR, Agarwal N, Bishai WR. Differential gene expression in response to exposure to antimycobacterial agents and other stress conditions among seven *Mycobacterium tuberculosis* whiB-like genes. *Antimicrob Agents Chemother* 2006;50:2836–2841.
- Flårdh K, Findlay KC, Chater KF. Association of early sporulation genes with suggested developmental decision points in *Streptomyces coelicolor* A3(2). *Microbiol* 1999;145:2229–2243.
- Dubnau E, Chan J, Mohan VP, Smith I. Responses of *Mycobacterium tuberculosis* to growth in the mouse lung. *Infect Immun* 2005;73:3754–3757.
- Gomez JE, Bishai WR. whmD is an essential *Mycobacterium* gene required for proper septation and cell division. *Proc Natl Acad Sci U S A* 2000;97:8554–8559.
- Kim T-H, Park J-S, Kim H-J, Kim Y, Kim P et al. The whcE gene of *Corynebacterium glutamicum* is important for survival following heat and oxidative stress. *Biochem Biophys Res Commun* 2005;337:757–764.
- Muttucumaru DGN, Roberts G, Hinds J, Stabler RA, Parish T. Gene expression profile of *Mycobacterium tuberculosis* in a non-replicating state. *Tuberculosis* 2004;84:239–246.
- Cole ST, Brosch R, Parkhill J, Garnier T, Churcher C et al. Deciphering the biology of *Mycobacterium tuberculosis* from the complete genome sequence. *Nature* 1998;393:537–544.
- Mulder NJ, Zappe H, Steyn LM. Characterization of a *Mycobacterium tuberculosis* homologue of the *Streptomyces coelicolor* whiB gene. *Tuber Lung Dis* 1999;79:299–308.
- Soliveri JA, Gomez J, Bishai WR, Chater KF. Multiple paralogous genes related to the *Streptomyces coelicolor* developmental regulatory gene whiB are present in *Streptomyces* and other actinomycetes. *Microbiol* 2000;146:333–343.
- Bush MJ. The actinobacterial WhiB-like (Wbl) family of transcription factors. *Mol Microbiol* 2018;110:663–676.
- Jakimowicz P, Cheesman MR, Bishai WR, Chater KF, Thomson AJ et al. Evidence that the *Streptomyces* developmental protein WhiD, a member of the WhiB family, binds a [4Fe-4S] cluster. *J Biol Chem* 2005;280:8309–8315.
- Alam MS, Garg SK, Agrawal P. Studies on structural and functional divergence among seven WhiB proteins of *Mycobacterium tuberculosis* H37Rv. *Febs J* 2009;276:76–93.
- Yan L, Zhang Q, Virolle M-J, Xu D. In conditions of over-expression, Wbl, a WhiB-like transcriptional regulator, has a positive impact on the weak antibiotic production of *Streptomyces lividans* TK24. *PLoS One* 2017;12:e0174781.
- Murarka P, Bagga T, Singh P, Rangra S, Srivastava P. Isolation and identification of a TetR family protein that regulates the biodesulfurization operon. *AMB Express* 2019;9:71.
- Singh P, Chachan S, Singhi D, Srivastava P. Isolation and molecular characterization of a stationary phase promoter useful for gene expression in *Gordonia*. *Gene* 2016;591:153–160.
- Russell DW, Sambrook J. Molecular cloning: a laboratory manual: cold Spring Harbor laboratory cold Spring harbor. *Cold Spring Harb Protoc* NY 2001.
- Singh P, Srivastava P. An improved protocol for electroporation in members of the genus *Gordonia*. *J Microbiol Methods* 2013;95:114–116.
- Singhi D, Jain A, Srivastava P. Localization of low copy number plasmid pRC4 in replicating rod and non-replicating cocci cells of *Rhodococcus erythropolis* PR4. *PLoS One* 2016;11:e0166491.
- Murarka P, Srivastava P. An improved method for the isolation and identification of unknown proteins that bind to known DNA sequences by affinity capture and mass spectrometry. *PLoS One* 2018;13:e0202602.
- Jaspers MC, Suske WA, Schmid A, Goslings DA, Kohler HP et al. HbpR, a new member of the XylR/DmpR subclass within the NtrC family of bacterial transcriptional activators, regulates expression of 2-hydroxybiphenyl metabolism in *Pseudomonas azelaica* HBP1. *J Bacteriol* 2000;182:405–417.
- Geertz M, Maerkl SJ. Experimental strategies for studying transcription factor-DNA binding specificities. *Brief Funct Genomics* 2010;9:362–373.
- Gordán R, Hartemink AJ, Bulyk ML. Distinguishing direct versus indirect transcription factor-DNA interactions. *Genome Res* 2009;19:2090–2100.
- Rybniiker J, Nowag A, van Gumpel E, Nissen N, Robinson N et al. Insights into the function of the WhiB-like protein of mycobacteriophage TM4—a transcriptional inhibitor of WhiB2. *Mol Microbiol* 2010;77:642–657.
- Davis NK, Chater KF. The *Streptomyces coelicolor* whiB gene encodes a small transcription factor-like protein dispensable for growth but essential for sporulation. *Mol Gen Genet* 1992;232:351–358.
- Bush MJ, Chandra G, Bibb MJ, Findlay KC, Buttner MJ. Genome-wide chromatin immunoprecipitation sequencing analysis shows that WhiB is a transcription factor that cocontrols its regulon with WhiA to initiate developmental cell division in *Streptomyces*. *mBio* 2016;7:e00523–16.
- Yamamoto K, Nishimura M, Kato D, Takeo M, Negoro S. Identification and characterization of another 4-nitrophenol degradation gene cluster, NPS, in *Rhodococcus* sp. strain PN1. *J Biosci Bioeng* 2011;111:687–694.

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